

Deepak Mallampati

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PROFESSIONAL SUMMARY

Applied AI engineer (MS CS, 2.5+ years) with production deployment experience across containerized AI services, CI/CD automation, and database infrastructure monitoring. Built FastAPI/Docker inference APIs with structured logging and load simulation for enterprise clients; designed Grafana/SQL observability dashboards for performance monitoring. Familiar with enterprise AI integration patterns and REST API production deployment. Open to EU work authorization discussions for the right opportunity.

CORE COMPETENCIES

Containerized AI Deployment (Docker) • CI/CD Pipeline Automation (Jenkins) • Infrastructure Monitoring (Grafana) • Enterprise REST API Integration • LLM Application Engineering • RAG Systems • Python / FastAPI / SQL • Enterprise Client Deployment

PROFESSIONAL EXPERIENCE

Progress Solutions Inc.

Jul 2025 - Present

Jr. AI/ML Engineer - Healthcare Technology, AI Consulting

- Packaged ML/LLM inference as **FastAPI/Flask RESTful services** with **Docker containerization**, structured logging, and load simulation; reduced post-deployment defects by **~30%**.
- Built RAG systems for enterprise knowledge retrieval; improved precision **~35%**.
- Developed agentic LLM workflows for multi-step enterprise queries; improved stability **~25%**.
- Built data preprocessing pipelines; raised dataset reliability above **98%**.

Energy Solutions International (Emerson)

Jun 2022 - Apr 2023

Database & DevOps Performance Engineer (Intern)

- Built **CI/CD pipelines with Jenkins** for schema updates and stored procedure deployments; reduced deployment errors **>30%**, improved release cycle time **~35-40%**.
- Designed **Grafana performance dashboards** using SQL Server DMVs to track long-running queries, deadlocks, CPU/I/O contention; cut incident recurrence **~25%**.
- Database maintenance (index rebuilds, partition-aware indexing, statistics updates) and RBAC implementation in compliance-sensitive environment.

Suvidha Foundation

Jan 2022 - Mar 2022

Machine Learning Engineer

- Flask-based API deployment; RAG document Q&A; transformer-based ML pipeline.

KEY PROJECTS

Agentic Claims Processing - Enterprise AI Integration

Multi-agent pipeline with schema-validated contracts, RAG-grounded regulatory validation, and production deployment. Demonstrates enterprise AI workflow integration at production scale.

Manga Lens - Multi-Provider AI Production Extension

Production Chrome extension integrating four AI providers. Multi-provider payload abstraction, error handling, 7-day cache. Demonstrates enterprise AI integration architecture and reliability engineering.

EDUCATION

Master of Science, Computer Science - Kent State University, Ohio, USA

Aug 2023 - May 2025

SKILLS

Languages: Python, TypeScript, SQL, JavaScript

Infra & DevOps: Docker, Jenkins CI/CD, Grafana, REST APIs, FastAPI, Flask

LLM & AI: RAG, agentic workflows, LangChain, LlamaIndex, prompt engineering

ML: PyTorch, scikit-learn, XGBoost, HuggingFace

Work Authorization: F-1 OPT (US-authorized); open to EU work authorization discussions